

Q1. Paper Analysis and Experimental Design

- a) The assumption about automated crawlers that the paper is examining is that automated web crawlers see essentially the same web that normal users can see. This matters because researchers use automated crawlers to study things such as privacy, security, and web behavior, so if a website is treating their crawler differently, then conclusions for their research are not going to accurately represent real user experiences.

- b) A research question I found comes from Section 1, the introduction of the paper, where it states, how do choices in browser configuration (BC) and network vantage point (VP) affect common privacy and security metrics?

An important result I found that ties a bit to the research question is found on Figure 5, around Section 3.2.2, where it states that browser configuration had a substantial effect on measurements of advertising and tracking traffic. The paper found that ~20% of consistently observed ad and tracker domains showed traffic that was strongly associated with the browser configuration being used.

Finally, for the limitation I found in Section 4.2, Limitations, the researchers admitted that they had resource constraints and therefore, their findings cannot be generalized to every residential, university, or cloud network.

- c) The bias score, as described in the paper, is the relative difference between two measurements using the difference of logarithms, which provides more descriptive power than a count difference and avoids extreme outliers when using ratios.

For an example, we can have $a = 8$ and $b = 2$. Then...

$$B(8,2) = R(\log_2 8/2) = R(\log_2 4) = R(2) = 2$$

The sign would be positive since a is larger than b , and the magnitude is 2 since a is $2^2=4$ times larger than b .

- d) A methodical lesson I will apply to question 2 is to change only one intended variable while keeping other variables as consistent as possible. For the User-Agent experiment, my independent variable will be the User-Agent header, and my measured outcomes will be the HTTP status code, page content, etc. My control variables will involve the URL, network connection, etc. An uncontrolled uncertainty could be that the website itself may change between requests due to advertisements or server load, so even if the responses differ, I cannot reach an absolute conclusion since I would not be sure that the User-Agent was the one that caused the difference.

Q2 Predictions

1. Default curl vs. Browser-like User-Agent

I predict that using a browser-like User-Agent will make the server return a more complete version of the page than a default curl. I will examine the status code,

response size, and whether an error page appears in the saved HTML. The server could treat the requests differently because the User-Agent identifies what type of client is making the request. If the default curl is identified as automated traffic, it could reduce content or receive an error or block page.

2. Default request vs. Different Accept-Language

I predict that changing the Accept-Language header will change some of the content returned by the server, like interface labels and possibly the language of review-related content. I will inspect the response hashes, response sizes, interface labels, dates, and any text between the two responses. The server could honor this header and return a localized version of the page based on the end user's language preference. The server could also ignore this header if the website controls language through another mechanism, like through the URL.

3. Initial Request vs. Request reusing saved cookies

I predict that reusing saved cookies will cause some page content to change compared to the initial request. I do think the preferences and related content will change. Cookies can store information about a previous visit; therefore, the cookie could represent the state of where a returning end user last was rather than showing content that would normally appear to a first-time visitor.

Q2. Controlled HTTP Experiment

Setup

URL: <https://www.yelp.com/biz/sushi-and-k-grill-fenton>

Date + Time: Tue Sep 22 12:22:38 CDT 2026

Curl Version: curl 8.1.1

Network Context: Home Wifi

Baseline

Command:

```
curl -L -D baseline-headers.txt \
```

```
-o baseline.html \
```

```
"https://www.yelp.com/biz/sushi-and-k-grill-fenton"
```

Timestamp: Tue Sep 22 12:49:37 CDT 2026

Final URL: <https://www.yelp.com/biz/sushi-and-k-grill-fenton>

Status code: 403 Forbidden

Content type: text/html; charset=UTF-8

Saved response body: baseline.html

Byte count: 776 bytes

SHA-256: ca02...3edd

Body classification: Block page, no review data was returned. The response asked the user to enable JS and disable any ad blocker and loaded a CAPTCHA script from captcha-delivery.com.

Browser-like User-Agent

Command:

```
curl -L -D useragent-headers.txt \  
-o useragent.html \  
-A "Mozilla/5.0" \  
"https://www.yelp.com/biz/sushi-and-k-grill-fenton"
```

Timestamp: Tue Sep 22 12:54:11 CDT 2026

Final URL: <https://www.yelp.com/biz/sushi-and-k-grill-fenton>

Status code: 403 Forbidden

Content type: text/html; charset=UTF-8

Saved response body: useragent.html

Byte count: 776 bytes

SHA-256: 0de3...acde

Body classification: Block page, no review data was returned. The response asked the user to enable JS and disable any ad blocker and loaded a CAPTCHA script from captcha-delivery.com.

Different Accept-Language

Command:

```
curl -L -D language-headers.txt \  
-o language.html \  
-H "Accept-Language: es-ES" \  
"https://www.yelp.com/biz/sushi-and-k-grill-fenton"
```

Timestamp: Tue Sep 22 12:56:39 CDT 2026

Final URL: <https://www.yelp.com/biz/sushi-and-k-grill-fenton>

Status code: 403 Forbidden

Content type: text/html; charset=UTF-8

Saved response body: language.html

Byte count: 776 bytes

SHA-256: 62b3...8219

Body classification: Block page, no review data was returned. The response asked the user to enable JS and disable any ad blocker and loaded a CAPTCHA script from captcha-delivery.com.

Cookie Reuse

Command:

```
curl -L -D cookies-headers.txt \
```

```
-o cookies.html \
```

```
-b cookies.txt \
```

```
"https://www.yelp.com/biz/sushi-and-k-grill-fenton"
```

Timestamp: Tue Sep 22 12:59:07 CDT 2026

Final URL: https://www.yelp.com/biz/sushi-and-k-grill-fenton

Status code: 403 Forbidden

Content type: text/html; charset=UTF-8

Saved response body: cookies.html

Byte count: 776 bytes

SHA-256: 7376...8ba9

Body classification: Block page, no review data was returned. The response asked the user to enable JS and disable any ad blocker and loaded a CAPTCHA script from captcha-delivery.com.

- a) For the Browser-Like User-Agent, my prediction was contradicted. I predicted that this request would make the server return a more complete version of the page than the baseline curl request. Yet, both the baseline request and the browser-like User-Agent request returned an HTTP 403 block page, the same response size of 776 bytes, and contained no review data. Therefore, changing the User-Agent did not result in a more complete page.
For the Different Accept-Language, there was not enough evidence to fully evaluate it. I predicted that changing the Accept-Language header would change interface labels or other localized content, but the request still returned an HTTP 403 block page as well, so I could not compare anything. Something of note is that the HTML language attribute did change from “en” to “es”, which shows that the server did respond in a way to the Spanish Accept-Language header.
Finally, for Cookie Reuse, my prediction was contradicted again. This is because I predicted that reusing saved cookies would cause some page content to change compared with the baseline request, but the cookie reuse request still returned the same HTTP 403 block page.
- b) A transport-level observation that I had was that all four requests, including the baseline, returned 403 status codes and text/html; charset=UTF-8. An application-level observation I had was that all of the responses contained a block page, and at least for the Accept-Language request, this request changed the HTML language attribute to Spanish. Still, these are not equivalent because the HTTP status and headers describe how the server responded at the protocol

level, while the application-level observation describes the actual content returned on the page.

- c) All of my bodies had different SHA-256 hashes, but none of them contained review data. The hashes still differed from one another because values inside the block pages changed and even the Accept-Language response also changed the HTML language attribute. This means that different hashes prove that only the response bodies were not the same, and that it cannot prove that any of the review data changed.
- d) A confounder in my experiment was that all four requests were made one after another at slightly different times. Because Yelp is a dynamic website and also uses automated bot protection, the server state or challenge information could have changed between requests. This means that some differences between the responses may not have been caused only by the User-Agent, Accept-Language, or cookies that I was changing. So, to make the experiment more reliable, I would repeat the same four controlled requests multiple times while keeping the time between requests as consistent as possible and still changing only the one intended variable. This would make it easier to determine whether any differences were actually caused by the request configuration instead of changes happening on Yelp's side.

Q3. Browser-Based Extraction and Comparison

- a) The path from HTTP response to browser DOM to selector to exported record starts with the response, as the data is initially returned by the server to the browser. The browser then parses said response and creates the DOM, which are the elements that are currently available on the webpage. JS can be used to modify the DOM after the HTTP response by adding, removing, or editing elements. Web Scraper specifically operates on this browser DOM using the selectors made in the sitemap. The scraper first selects each review container and then extracts the specified information for the homework from that page. These extracted values are then combined into records and can be exported to CSV.
- b) The five records I will compare are from Melanie S, who gave a 5 star rating on May 3rd, 2026, Christina C, who gave a 3 star rating on Jul 23rd, 2026, Maggie H, who gave a 5 star rating on Sep 11th, 2026, Trey C, who gave a 5 star rating on Aug 31st, 2026, and Eric B, who gave a 5 star rating on Jul 29th, 2026. All classifications are missing when compared to my saved curl responses because all responses gave me HTTP 403 codes and did not contain any of the review records.
- c) The most consequential mismatch, in general, was that the browser contained review data that the curl response could not grab. A possible explanation could

come from Yelp's own automated bot protection which recognized the curl request as automated traffic and forcibly returned a block page. A second explanation could be that some of the review information is loaded by JS after the browser receives its initial HTTP response; therefore, any of the curl responses would not have been able to receive any data. An additional test I could run would be to disable JS in the browser and reload the Yelp page. If the reviews disappear, that would support the explanation that JS loads or inserts the review data. If the reviews remain visible without JS, that would provide more support for the explanation that Yelp returned different initial content to the browser than it returned to curl.

- d) Five plausible-looking rows and a screenshot are not proof that the extraction is correct because if I messed up the selectors on Web Scraper, it could produce data that skips reviews, duplicates reviews, or even associates data with a different reviewer. A screenshot is also limited because that is only showing a small portion of the extraction process, even with the screenshots I provided, it could not show all of the extraction data which means that the accuracy of every exported record cannot be confirmed.